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Practicum CFD 2026-2027

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Practicum 3
deadline: Monday, September 28th, 17:00

- Working in pairs (groups of max. 2 students) is allowed.
- Each practicum is closed with an oral discussion supported by a (short) written report of the findings. Make the report by pen/paper (readable) or with LaTeX, Word, etc.
- Always use Python or Matlab/Octave for the programs that you develop.
- Save your programs in separate files, such that it is easy to demonstrate them.
- The use of AI tools such as ChatGPT is not allowed!
- Be sure that you have discussed the exercises and programs before the deadline!

Exercise 1 – JOR on the computer

Description After discretization of a (system of) p.d.e.(s) usually a large linear system $Ax = b$ is obtained. Solving such a large system with a direct method is expensive with respect to both memory and CPU time. Therefore, such systems are often solved with an iterative method.

As in Practicum 1 and 2, we consider the inhomogeneous equation on $0 \leq x \leq 1$

$$\frac{\mathrm{d}\phi}{\mathrm{d}x} - k\frac{\mathrm{d}^2\phi}{\mathrm{d}x^2} = \begin{cases} 2 - 5x & , \quad 0 \leq x \leq 0.4 \\ 0 & , \quad 0.4 < x \leq 1 \end{cases}$$

with $\phi(0) = 0$, $\phi(1) = 1$. The Matlab file `cfd_3.m` discretizes this equation with a number of methods and solves the resulting linear systems iteratively with the JOR (or SOR) method. As stopping criterion in these methods $\| \phi^{n+1} - \phi^n \|_\infty \leq 10^{-5}$ is used, with n the counter of the iterative steps. Hence, the difference between two consecutive solutions must be sufficiently small.

Two types of grid are used: uniform grids and non-uniform (stretched) grids. The stretched grid possesses $N/2$ equal grid cells between 0 and $1 - 5k$, and $N/2$ equal grid cells between $1 - 5k$ and 1. Several types of discretization are employed. On the uniform grid both upwind and central discretization are used. On the stretched grid upwind, central A and central B are used.

Required files: `cfd_3.m`, `plotev.m`, `plotsol.m`, `ns_iterative.m`, `ns_jor.m`, `ns_sor.m`, `ns_exact.m`.

Typing the command `cfd_3` in Matlab starts a menu in which the choice of the grid, the discretization method, the iterative method and the value of the relaxation factor ω can be given. Then, with the menu a program can be started which solves the convection-diffusion equation in the prescribed way.

The results produced by the program consist of the eigenvalues of the Jacobi matrix, a plot of the iteration error (this error is written to the screen as well), the total number of iterations and a plot showing the continuous solution, the discrete solution computed with a direct solver and the discrete solution computed with the iterative method (provided the iterative method did converge).

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Questions to be solved on the computer

Compute the solution for $k = 0.01$. As a default, the program uses $N = 10$ grid cells.

(a) Select the **uniform** grid. Use first the **upwind** and then the **central** method. Determine numerically whether the Jacobi method (i.e. the JOR method with $\omega = 1$) converges or not. Put the results in the table below.

uniform grid	Jacobi method	
	convergent?	# iterations
upwind	yes	25
central	NO	465

$1.39e^{-5}$ em INF

(1) Use the eigenvalues of the Jacobi matrix to explain the results.

(b) Select the **uniform** grid. Use first the **upwind** and then the **central** method. Use various (suitably chosen) values of ω and determine numerically for each value of ω the number of iterations needed by the JOR method. Determine in this way the optimal and maximal relaxation factor (ω_{opt} and $\omega_{\max}$) for JOR. Put the results in the table below.

uniform grid	JOR method			
	ω_{opt}	# it.	$\omega_{\max}$	# it.
upwind	1	25	1.31	11411
central	0.0451	317	0.088	$\sim 10^5$

(1) Which systems are solved better by the JOR method, those resulting from upwind or those resulting from central discretization? Why?

(c) Select the **stretched** grid. Use first the **upwind**, then the **central A** and finally the **central B** method. Determine numerically whether the Jacobi method ($\omega = 1$) converges or not. Put the results in the table below.

stretched grid	Jacobi method	
	convergent?	# iterations
upwind	yes	83
central A	NO	359
central B	NO	358

(1) Use the eigenvalues of the Jacobi matrix to explain the results.

(d) Select the **stretched** grid. Use first the **upwind**, then the **central A** and finally the **central B** method. Use various (suitably chosen) values of ω and determine numerically for each value of ω the number of iterations needed by the JOR method. Determine in this way the optimal and maximal relaxation factor, ω_{opt} and $\omega_{\max}$ for the JOR method. Put the results in the table below.

stretched grid	JOR method			
	ω_{opt}	# it.	$\omega_{\max}$	# it.
upwind	0.99	79	1.057	23108
central A	0.02501	1191	0.032	6154
central B	0.0077	1751	0.0077	1751

(1) In case of the central B method, compare the discrete solutions obtained using JOR and the exact solver. Does JOR really converge? Use the eigenvalues of the Jacobi matrix to explain this result.
(2) Which systems are solved better by the JOR method, those resulting from the upwind, the central A or the central B discretization?

1(a)

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eigenvalues

Central	upwind
$0, \pm 1.5i, \pm 2.8i, \pm 3.9i, -4.6i$	$\pm 0.52, \pm 0.44; \pm 0.32, \pm 0.17, 0$

The eigenvalues of the Jacobi matrix are fully imaginary for the central discretization scheme, while fully real for the upwind.

the spectral radius for the central discretization is

$$\rho_{JC} = \max |\mu_{JC}| = 4.6 > 1$$

thus the method does not indeed converge.
Additionally from (A.18a), the JOR eigenvalues satisfy

$$\mu_{JOR,C} = 1 + \omega(\mu_{JC} - 1)$$

the notes (Thm A.2.3, Eq. A.19) give, for a largest purely imaginary Jacobi eigenvalue

$$\mu_{JC} = i\mu_I \Rightarrow \omega_{max} \leq \frac{2}{1+\mu_I^2} \approx 0.09$$

Meaning, the method needs $0 < \omega \leq 0.09$ for converging, which is indeed the case if we run with $\omega = 0.08$ (it converge within 1010 iterations). Following the notes still, the optimal converging factor would be $\omega \approx 0.0451$ since $\rho_{opt} \approx 0.977$. Indeed in this case it converge into 317 steps.

the upwind method converge, similarly because

1. the spectral radius is smaller than 1
2. From A.2.1 the upwind method has a M-matrix, meaning, the matrix is irreducible and weakly diagonally dominant $\Rightarrow$ the Jacobi converges.

Solving following the notes, the ω_{opt} is 1. the ω is bound by 4/3.

1(b) Before, from the formula from the notes we derived the ω_{max} for each method. Testing out them we found that both methods diverge actually at those values but they converge when really close to it.

to evaluate which one is better consider that JOR iteration requires $\rho(I - \omega D^{-1}A)$ to be small, the smaller the faster the convergence rate. So, Although from this at ω_{opt} for central the ρ is approx 0.977 which is really close to 1 hence it is expected to have a slow convergence rate, in particular in respect to upwind that has a ρ of 0.5.

taking upwind the error behaves as

$$e^{(n)} \approx \rho^n e^{(0)} = (0.5)^n e^{(0)}$$

$0.977 \gg 0.5$

instead of central

$$e^{(n)} \approx \rho^n e^{(0)} = (0.977)^n e^{(0)}$$

As done before, the most restrictive eigenvalue for upwind is -0.5. Implies $\omega < 4/3 \approx 1.33$. Numerically, $\omega_{max} \approx 1.31$ which agrees with this one

$$\omega \to \omega_{max} \Rightarrow \rho(\cdot) \to 1$$

Similar reasoning for central. In conclusion, upwind is better because ρ is considerably smaller than central.

1(c) Similarly to 1(a),

upwind	$\pm 0.89, \pm 0.55; \pm 0.34; \pm 0.139; 0$
A	$\pm 0.48; \pm 0.79; \pm 3.17i; \pm 7.68i; 0$
B	$\pm 0.56; \pm 1.0015; \pm 2.9i; \pm 7.6i; 0$

Same reasoning as 1a ρ of methods A & B is bigger than 1.

1(d) Similar reasoning of 1b:

	ω_{max} <	ω_{opt} <	ρ	
upwind	1.058	1	0.89	
A	0.0334	0.017	0.992	→ For Method A the opt is $\omega = 0.02501$.

For B is not really possible to find optimal because we have $\rho > 1$ (due to 1.0015) $\forall \omega \geq 0$. the methods diverge asymptotically always.

Method B at $\omega = 0.017$ then it initially decrease to $\approx 2 \times 10^{-5}$ (what it would be the "theoretical one") so almost to the stopping criterion. After that minimum it goes back up but really slowly. $\omega = 0.016, 0.015, 0.014, 0.013, \dots, 0.01$

As we decrease also the minimum will decrease

Really close to 10^{-5}

Until 0.0077. this optimal is also the ω_{max}, because in the end since the method is always divergence asymptotically then in practical setting we can find a value ω that sufficiently delay the growth of the unstable eigenvalues, which is what happen with $\omega = 0.017 \to 0.0078$. In those cases the minimum is not sufficient to stop the algorithm.

Indeed, when also looking at those that converge, actually the solution did not converged at all. In other words, although $\omega < \omega_{max}$ and we do not really have convergence mainly due to the $\mu = 1.0015$

Major problem

Compared to the plots, the method A is the one yielding the best result overall. this is in line with what discussed in Practical 2.

Exercise 2 – Analysis of Jacobi

(a) Consider the uniform grid, with $k = 0.01$ and $N = 10$.
(1) Check whether the coefficient matrices for upwind and central discretisation are diagonally dominant.
(2) Does this explain the results in 1(a)?

(b) Consider the stretched grid, with $k = 0.01$ and $N = 10$.
(1) Check whether the coefficient matrices for upwind, central A and central B are diagonally dominant.
(2) Does this explain the results in 1(c)?

Exercise 3– Further analysis of JOR

(a) In Section A.2.1 for a central discretization on the uniform grid the Jacobi matrix A_J and its eigenvalues μ_J are derived

$$A_J = \text{diag}[\frac{P}{4} + \frac{1}{2}, 0, -\frac{P}{4} + \frac{1}{2}], \quad \mu_J = \frac{1}{2}\sqrt{4 - P^2}\cos(\frac{i\pi}{N}), \quad i = 1, \cdots, N-1,$$

where N is the number of grid cells and P is the mesh-Péclet number.
(1) Derive in a similar way for the **upwind** discretization on the **uniform** grid the eigenvalues μ_J of the Jacobi matrix. Hint: use Lemma A.2.10.
(2) Show that these eigenvalues are real and between -1 and 1 for all P values.

(b) For a central discretization on the uniform grid the eigenvalues of the Jacobi matrix are pure imaginary when $P > 2$. Let the largest pure imaginary eigenvalue be given by $\mu_J = i\mu_I$, with μ_I real. The optimal and maximal relaxation factor for the JOR method are given by (cf. Section A.2.1)

$$\omega_{\text{JOR,opt}} = \frac{1}{1 + \mu_I^2}, \quad \omega_{\text{JOR,max}} = \frac{2}{1 + \mu_I^2}.$$

For an **upwind** discretization on the **uniform** grid the eigenvalues of the Jacobi matrix are real, see (a). Let the largest real eigenvalues be given by $\mu_J = \mu_R$ and $\mu_J = -\mu_R$, with μ_R real.
(1) Derive the optimal and maximal relaxation factors for JOR as a function of μ_R. Hint: the eigenvalues μ_{JOR} of the iteration matrix of JOR are related to the eigenvalues μ_J of the Jacobi matrix by $\mu_{\text{JOR}} = 1 + \omega(\mu_J - 1)$, and the JOR method converges when the eigenvalues μ_{JOR} lie within the unit circle.

(c) In Matlab Exercise (1), we used $k = 0.01$ and $N = 10$.
(1) Use 3(a) and 3(b) to determine for these values, for the JOR method, the optimal and maximal relaxation factors in case of discretization on the **uniform** grid with respectively the **upwind** and **central** method.
(2) Compare these theoretical results with the numerical results from 1(b).

(d) Consider **upwind** discretization on the **stretched** grid (again k=0.01, N=10).
(1) Use 3(b) and the numerical values of the eigenvalues μ_J of the Jacobi matrix, computed as byproduct by `cfd_3` in 1(d), to determine for JOR the theoretical values of the optimal and maximal relaxation factor.
(2) Compare these theoretical results with the numerical results from 1(d), for the **upwind** discretization on the **stretched** grid.
(3) Why, in general, is it difficult to compute the theoretical optimal and maximal relaxation factor in case of **central A** or **central B** discretization on the **stretched** grid?

2(a) In upwind we have

$$a_{ii} = 2\frac{\kappa}{h^2} + \frac{1}{h} = 2+10 = 12$$

$$a_{i,i-1} = -\left(\frac{\kappa}{h^2} + \frac{1}{h}\right) = -11 \qquad a_{i,i+1} = -\frac{\kappa}{h^2} = -1$$

Hence $|a_{ii}| = 12$, $|a_{i,i-1}| + |a_{i,i+1}| = 12$

Therefore, the matrix is weakly diagonally dominant => Jacobi converges By theorem A.2.1

Central, instead, is not since

$$a_{ii} = 2\frac{\kappa}{h^2} = 2, \quad a_{i,i-1} = -\frac{\kappa}{h^2} - \frac{1}{2h} = -6, \quad a_{i,i+1} = -\frac{\kappa}{h^2} + \frac{1}{2h} = 4$$

meaning $|a_{ii}| = 2$, $|a_{i,i-1}| + |a_{i,i+1}| = 10$.

Which explain results in 1a. the upwind method is weakly diagonally dominant, and indeed the Jacobi have $\rho(J) = 0.5 < 1$ and SOR perform well.
the central, instead, is not diagonally dominant and the $\rho(J) > 1$ meaning the Jacobi diverges. SOR then only converges after strong over-relaxation ($\omega \approx 0.045$)

2(b) As said before, the upwind produces a positive diagonally entries and non negative off-diagonal entries yielding a M-structure. Therefore, the matrix remain weakly diagonal dominant.

Central A: Using formulas 1.18,

$$a_{i,i-1} = -\frac{1}{h_- + h_+} - \frac{2\kappa}{(h_- + h_+)h_-} \rightsquigarrow 0$$

$$a_{ii} = \frac{2\kappa}{h_- + h_+}\left(\frac{1}{h_-} + \frac{1}{h_+}\right) \quad 0$$

$$a_{i,i+1} = \frac{1}{h_- + h_+} - \frac{2\kappa}{(h_- + h_+)h_+} \quad 0$$

Since $\kappa = 0.01$ then

$$|a_{ii}| < |a_{i,i-1}| + |a_{i,i+1}|$$

meaning method A is not diagonally dominant.

Similarly for method B

$$a_{ii} = \frac{h_+ - h_-}{h_- h_+} + \frac{2\kappa}{h_- + h_+}\left(\frac{1}{h_-} + \frac{1}{h_+}\right) \qquad a_{i,i-1} = -\frac{h_+}{h_-(h_- + h_+)} - \frac{2\kappa}{(h_- + h_+)h_-}$$

$$a_{i,i+1} = \frac{h_-}{h_+(h_- + h_+)} - \frac{2\mu}{(h_- + h_+)h_+}$$

In the a_{ii} we have an extra $\frac{h_+ - h_-}{h_- h_+}$ term, which can be positive or negative depending on the chosen grid. In other words, some rows may become poorly conditioned losing the needed π-structure.

In other words also notice B is not diagonally dominant. This explains the previous results.

3(a) For the upwind on an uniform grid it is possible to rewrite the coefficient matrix

$$A = \text{DIAG}[-1,\ P+2,\ -(P+1)]$$

meaning the Jacobi is

$$J = \text{DIAG}\left[\underbrace{\frac{P+1}{P+2}}_{a}, 0, \underbrace{\frac{1}{P+2}}_{c}\right]$$

by lemma A.2.10 the eigenvalues are

$$\mu = 2\sqrt{ac}\cos\left(\frac{i\pi}{N}\right) \quad i = 1, \dots, N-1$$

As seen in previous section, $\forall P \geq 0$ $a, c > 0$ meaning μ is real. Since

$$\left|\cos\left(\frac{i\pi}{N}\right)\right| \leq 1$$

then it is suffice to show

$$\frac{2\sqrt{P+1}}{P+2} \leq 1 \implies 4(P+1) \leq (P+2)^2$$
$$\implies 0 \leq P^2 \quad \forall P$$

meaning

$$|\mu| \leq \frac{2\sqrt{P+1}}{P+2} \leq 1 \implies -1 \leq \mu \leq 1 \quad \forall P.$$

3(b) For the upwind discretization, the Jacobi eigenvalues are real & symmetric

$$\mu_J = \pm \mu_R \quad 0 < \mu_R < 1$$

From the SOR eigenvalue relation

$$\mu_{SOR} = 1 + \omega(\mu_J - 1) \implies |\mu_{SOR}| < 1 \quad \forall \mu_J$$

to determine the SOR eigenvalues corresponding to the extreme Jacobi eigenvalues we start from the above relation

$$\mu_+ = 1 + \omega(\mu_R - 1) = 1 - \omega(1 - \mu_R) \text{ where } \mu_J = \mu_R$$

Similarly for $\mu_S = -\mu_R$

$$\mu_- = 1 - \omega(1 + \mu_R)$$

Since all Jacobi eigenvalues lies within $-\mu_R, \mu_R$ these two determine the spectral radius:

$$\rho_{JOR} = \max(|\mu_+|, |\mu_-|)$$

the optimal relaxation factor minimize the spectral radius, meaning as ω increases

$$\mu_+ = 1 - \omega(1 - \mu_R)$$

moves slowly to the left while

$$\mu_- = 1 - \omega(1 + \mu_R)$$

moves faster to the left.

$$\left.\begin{aligned} \partial_\omega \mu_+ &= -(1 - \mu_R) \\ \partial_\omega \mu_- &= -(1 + \mu_R) \end{aligned}\right\} \quad \mu_R > 0 \Rightarrow 1 + \mu_R > 1 - \mu_R$$

$$\Rightarrow |\partial_\omega \mu_-| > |\partial_\omega \mu_+|$$

the optimal happen when both extreme eigenvalues have equal magnitude

$$|\mu_-| = |\mu_+|$$

meaning at the optimal one eigenvalue is positive while the other is negative

$$1 - \omega(1 - \mu_R) = -(1 - \omega(1 + \mu_R))$$

$$\Leftrightarrow \quad 1 - \omega + \omega\mu_R = -1 + \omega + \omega\mu_R$$

$$\Leftrightarrow \quad 1 - \omega = -1 + \omega$$

$$\Leftrightarrow \quad 2 = 2\omega$$

$$\Leftrightarrow \quad \boxed{\omega_{JOR,opt} = 1}$$

to find the optimal spectral we substitute $\omega = 1$ giving

$$\mu_+ = \mu_R \quad \& \quad \mu_- = -\mu_R$$

$$\Rightarrow \rho_{JOR,opt} = \mu_R$$

the maximum relaxation factor is the largest value of ω for which all eigenvalues remain inside the unit circle.

In this case, the restrictive eigenvalue is

$$\mu_- = 1 - \omega(1 + \mu_R)$$

Because it moves faster towards -1.

the convergence condition

$$|\mu_-| < 1 \Rightarrow -1 < 1 - \omega(1 + \mu_R) < 1$$

$$\Rightarrow -\omega(1 + \mu_R) < 0 \quad \forall \omega > 0$$

$$\& \quad 1 - \omega(1 + \mu_R) > -1$$

$$\Rightarrow 2 > \omega(1 + \mu_R)$$

$$\Rightarrow \boxed{\omega_{JOR,max} = \frac{2}{1 + \mu_R}}$$

the optimal & maximum relaxation factors are

$$\omega_{JOR,opt} = 1 \quad \& \quad \omega_{JOR,max} = \frac{2}{1 + \mu_R}$$

3(c) In this configuration, the Peclet number is $P=10$.

From 3(a), The Jacobi eigenvalues are

$$\mu_J = \frac{2\sqrt{11}}{12}\cos\left(\frac{\pi}{10}\right) \approx 0.5257$$

From 3(b)

$$\omega_{JOR,opt} = 1 \quad \& \quad \omega_{JOR,max} = \frac{2}{1+\mu_R} = 1.311$$

Which is in agreement with 1(b)!

For central discretization, $P>2$ which is the case and

$$\mu_J = i\mu_I \quad \text{with} \quad \mu_I = \sqrt{24}\cos\left(\frac{\pi}{10}\right) \approx 4.659$$

using the formula from A.2.1

$$\omega_{JOR,opt} = \frac{1}{1+\mu_I^2} = 0.0440 \quad \& \quad \omega_{JOR,max} = \frac{2}{1+\mu_I^2} = 0.0880$$

Which is in agreement with 1(b). We were close for the optimal. Indeed, running with $\omega_{opt} = 0.044$ we get 311 iterations!

3(d) In 1(d) the dominant eigenvalues are $\mu_R = \pm 0.8918$. From 3(b) $\omega_{opt} = 1$ & $\omega_{max} = \frac{2}{1+\mu_R}$ meaning

$$\omega_{max} = 1.0572 \quad \& \quad \omega_{opt} = 1$$

We have a really good agreement with the numerical results. As seen earlier on, as $\omega \to \omega_{max}$ $\rho(J_{OR}) \to -1$ meaning the convergence become very slowly As seen in the # it incorrect.

Why it is hard for central A&B? Well, the upwind has a clean eigenvalues structure with $\mu_J \in [-\mu_R, \mu_R]$ which allows us, as done before, to reach a clean simple formula. For stretched-grid method A the spectrum is mix of purely real and purely imaginary eigenvalues. Similarly for method B.

In other words, the spectrum radius is

$$\rho(\omega) = \max_i |1 + \omega(\mu_{J,i} - 1)|$$

Now more than one eigenvalues can define the maximum $\Leftrightarrow$ As ω changes different eigenvalues may become dominant. This is harder to find an opt & max for $\rho(\omega)$!

As seen before, method B, in this case, has even one eigenvalue 1.0015 which means the method is theoretically divergent $\forall w \geq 0$, although it may exhibit apparent initial convergence before the unstable eigenvalue become dominant.